

Lecture 8: Conditioning, Bayes' Rule

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1 Conditioning

1.1 Conditioning a Random Variable on an Event

i Definition: conditional PDF given an event

The conditional PDF $f_{X|A}$ of a continuous random variable X , given an event A with $P(A) > 0$ satisfies

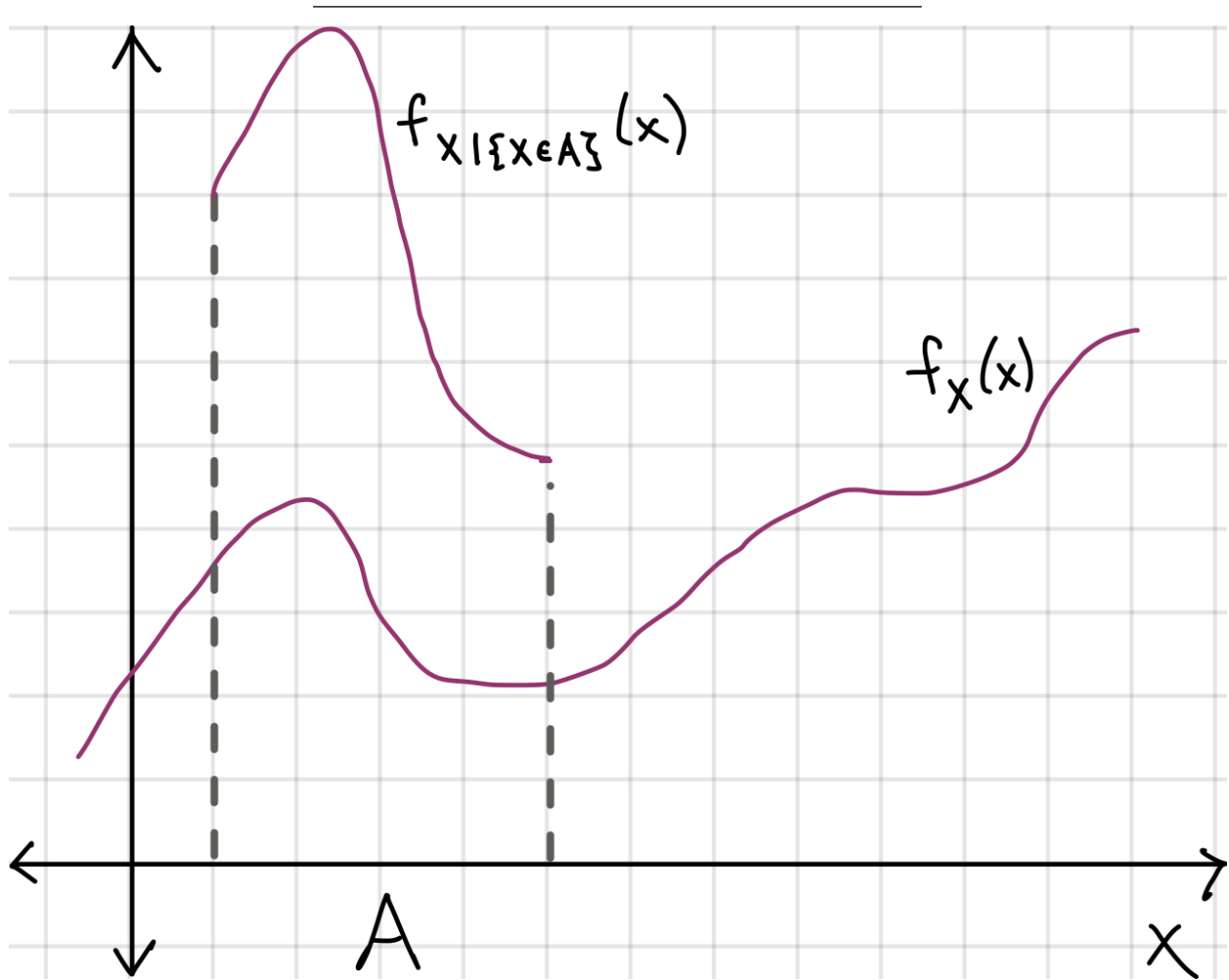
$$P(X \in B | A) = \int_B f_{X|A}(x) dx.$$

If $A \subset \mathbb{R}$ with $P(X \in A) > 0$, then

$$f_{X|\{X \in A\}}(x) = \frac{f_X(x)}{P(X \in A)} \quad \text{for } x \in A,$$

and $f_{X|\{X \in A\}}(x) = 0$ otherwise.

Going from f_X to $f_{X|\{X \in A\}}$ zeroes out the mass/density outside A , and rescales what remains so it once again integrates (or sums) to 1.



1.1.1 Example: Memorylessness of Exponential RV

Let $T \sim \text{Exponential}(\lambda)$ model a lightbulb's lifetime, and condition on the event $A = \{T > s\}$: the bulb has already survived s hours.

$$P(A) = P(T > s) = \int_s^\infty \lambda e^{-\lambda t} dt = e^{-\lambda s}.$$

For $t \geq s$,

$$f_{T|T>s}(t) = \frac{f_T(t)}{P(T > s)} = \frac{\lambda e^{-\lambda t}}{e^{-\lambda s}} = \lambda e^{-\lambda(t-s)}.$$

💡 Lightbulb interpretation

Substituting $\tau = t - s$ (additional lifetime past time s), the density of τ given $T > s$ is $\lambda e^{-\lambda \tau}$ – still an $\text{Exponential}(\lambda)$ density. A bulb that has already burned for s hours is, from this moment on, statistically identical to a brand-new bulb.

1.1.2 Law of Total Probability

Just as a PMF can be decomposed over a partition, so can the PDF of a random variable:

i Theorem: law of total probability, RV form

If $A_1, \dots, A_n$ partition the sample space with $P(A_i) > 0$ for all i , then

$$f_X(x) = \sum_{i=1}^n P(A_i) f_{X|A_i}(x).$$

1.1.3 Example: A Population Mixture of Two Gaussians

A population's heights come from one of two subpopulations, each with its own Gaussian distribution:

$$X | M \sim N(70, 3^2), \quad X | F \sim N(62, 2.5^2).$$

By the law of total probability (with the two-event partition $\{M, F\}$):

$$f_X(x) = P(M) f_{X|M}(x) + P(F) f_{X|F}(x) = \frac{1}{2} \phi_{70,3}(x) + \frac{1}{2} \phi_{62,2.5}(x).$$

⚠ A mixture of Gaussians is *not*, in general, Gaussian

f_X above is not itself of the form $\phi_{\mu,\sigma}$ for any μ, σ — averaging two Gaussian densities does not give a Gaussian density. Second, bimodality isn't automatic: if the two subpopulation means are close relative to their spreads (e.g. 70 vs. 64 with the same 3 and 2.5), the same recipe produces a single-peaked f_X . Learning these Gaussian components is known as *Gaussian mixture models* in statistical and machine learning.

```
# Gaussian mixture: bimodal vs. unimodal, depending on separation
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm
from scipy import integrate

def mixture_pdf(x, p, mu1, s1, mu2, s2):
    return p * norm.pdf(x, mu1, s1) + (1 - p) * norm.pdf(x, mu2, s2)

# Case 1: well-separated means -> bimodal
p, mu1, s1, mu2, s2 = 0.5, 70, 3, 62, 2.5

# Case 2: same spreads, closer means -> unimodal
mu1_close, mu2_close = 70, 64

xs = np.linspace(45, 90, 2000)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5), sharey=True)

# --- Panel 1: bimodal case ---
axes[0].plot(xs, norm.pdf(xs, mu1, s1), color="0.6", ls="--", label=r"$f_{X|M}$")
```

```

axes[0].plot(xs, norm.pdf(xs, mu2, s2), color="0.6", ls=":", label=r"$f_{X\mid F}$")
axes[0].plot(xs, mixture_pdf(xs, p, mu1, s1, mu2, s2), color="C0", lw=2, label=r"$f_X$ (mixture)")
axes[0].set_title(f"Means {mu2} and {mu1}: bimodal")
axes[0].legend(fontsize=8)
axes[0].set_xlabel("x")

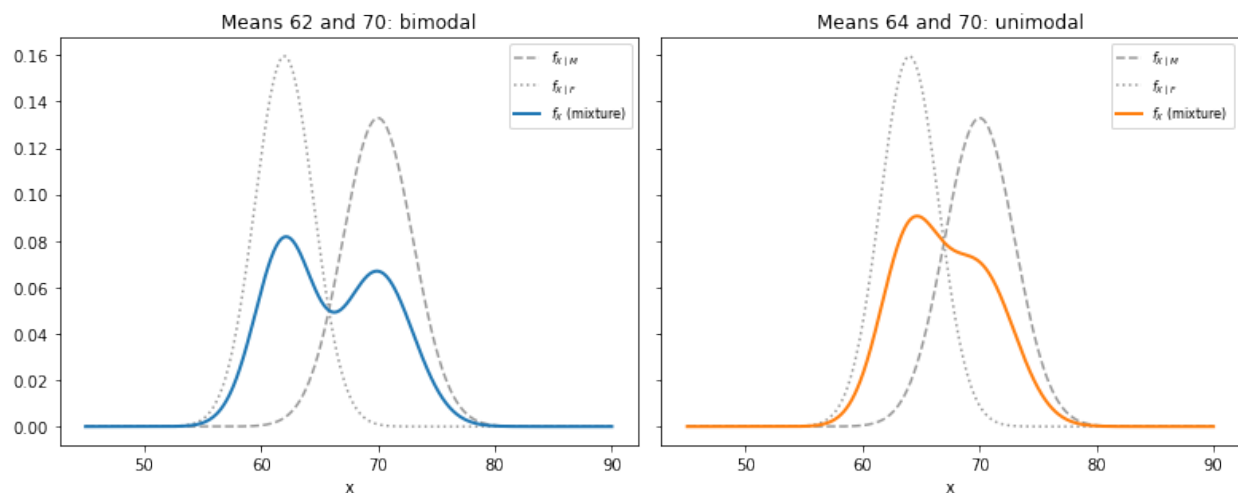
# --- Panel 2: unimodal case ---
axes[1].plot(xs, norm.pdf(xs, mu1_close, s1), color="0.6", ls="--", label=r"$f_{X\mid M}$")
axes[1].plot(xs, norm.pdf(xs, mu2_close, s2), color="0.6", ls=":", label=r"$f_{X\mid F}$")
axes[1].plot(xs, mixture_pdf(xs, p, mu1_close, s1, mu2_close, s2), color="C1", lw=2, label=r"$f_X$ (mixture)")
axes[1].set_title(f"Means {mu2_close} and {mu1_close}: unimodal")
axes[1].legend(fontsize=8)
axes[1].set_xlabel("x")

plt.tight_layout()
plt.show()

# --- Numerical checks ---
# 1) mixture integrates to 1
check1 = integrate.quad(lambda x: mixture_pdf(x, p, mu1, s1, mu2, s2), -np.inf, np.inf)[0]
check2 = integrate.quad(lambda x: mixture_pdf(x, p, mu1_close, s1, mu2_close, s2), -np.inf, np.inf)[0]

# 2) count local maxima (peaks) to confirm bimodal vs. unimodal numerically
def count_modes(p, mu1, s1, mu2, s2, lo=45, hi=90, n=4000):
    grid = np.linspace(lo, hi, n)
    vals = mixture_pdf(grid, p, mu1, s1, mu2, s2)
    d = np.diff(vals)
    peaks = grid[1:-1][(d[:-1] > 0) & (d[1:] < 0)]
    return peaks

```



1.2 Conditioning One RV on Another

i Definition: conditional PDF given another RV

For jointly continuous X, Y with joint PDF $f_{X,Y}(x, y)$ and y with $f_Y(y) > 0$,

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

For each fixed y , we take the *slice* of the joint density along $Y = y$ and rescale it (divide by $f_Y(y)$) so it integrates to 1 in x .

1.2.1 Example: Dartboard

Let (X, Y) be uniform on the unit disk (a dart thrown uniformly at a circular board of radius 1):

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{\pi}, & x^2 + y^2 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Recall that this has a *non-uniform* marginal, $f_X(x) = \frac{2\sqrt{1-x^2}}{\pi}$ for $|x| \leq 1$.

For $|x| < 1$, the conditional PDF is

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{1/\pi}{2\sqrt{1-x^2}/\pi} = \frac{1}{2\sqrt{1-x^2}}, \quad |y| < \sqrt{1-x^2}.$$

💡 Physical interpretation

Given only the dart's horizontal coordinate $X = x$, the possible vertical landing spots are the **chord** of the disk at that x – and Y is *uniform* over that chord, even though the joint density was uniform over the whole disk. Conditioning “slices” the flat joint density and renormalizes the slice; the slice itself is still flat, so it's uniform on its (shorter) interval.

1.3 Conditional Expectation

i Definition

Let X and Y be jointly continuous random variables, and let A be an event with $P(A) > 0$, then the conditional expectation is

$$E[X | A] = \int_{-\infty}^{\infty} x f_{X|A}(x) dx,$$
$$E[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx.$$

💡 LOTUS still holds

$$E[g(X) \mid Y = y] = \int g(x) f_{X|Y}(x \mid y) dx.$$

Conditioning just replaces the unconditional (marginal) distribution with the conditional one everywhere in the LOTUS formula.

📖 Theorem: total expectation theorem

Let $A_1, \dots, A_n$ form a partition of the sample space, and assume that $P(A_i) > 0$ for all i . Then,

$$E[X] = \sum_i P(A_i) E[X \mid A_i].$$

Similarly,

$$E[X] = E[E[X \mid Y]] = \int_{-\infty}^{\infty} E[X \mid Y = y] f_Y(y) dy.$$

1.3.1 Example: Dartboard, continued

By the symmetry $f_{Y|X}(y \mid x) = f_{Y|X}(-y \mid x)$ established previously,

$$E[Y \mid X = x] = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} y \cdot \frac{1}{2\sqrt{1-x^2}} dy = 0 \quad \text{for every } x.$$

By the total expectation theorem (RV form),

$$E[Y] = E[E[Y \mid X]] = \int_{-1}^1 0 \cdot f_X(x) dx = 0,$$

consistent with computing $E[Y]$ directly from the (also symmetric) marginal f_Y .

1.4 Independence

📖 Definition

Continuous X, Y are **independent** if

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \quad \text{for all } x, y,$$

equivalently $f_{X|Y}(x \mid y) = f_X(x)$ whenever $f_Y(y) > 0$.

💡 Two facts carried over from discrete case

Product rule for expectation. If X, Y independent, $E[XY] = E[X] E[Y]$ (the joint density factors, so the double integral in two-variable LOTUS splits into a product of two single integrals).

Linearity of variance. If X, Y independent, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

1.4.1 Example: Joint PDF of Independent Gaussian RVs

If $X_1, \dots, X_n$ are independent with $X_i \sim N(\mu_i, \sigma_i^2)$, their joint density is simply the product of the marginals:

$$\begin{aligned} f(x_1, \dots, x_n) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right) \\ &= \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)\right), \end{aligned}$$

where $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$.

i Definition: isotropic

If, further, every $\sigma_i = \sigma$, then $\Sigma = \sigma^2 I$ and f depends on $\mathbf{x}$ only through $\|\mathbf{x} - \mu\|^2$ – the density looks the **same in every direction**. Such a distribution is called **isotropic**. Its level sets (contours of equal density) are circles (or spheres, in higher dimension) rather than general ellipses.

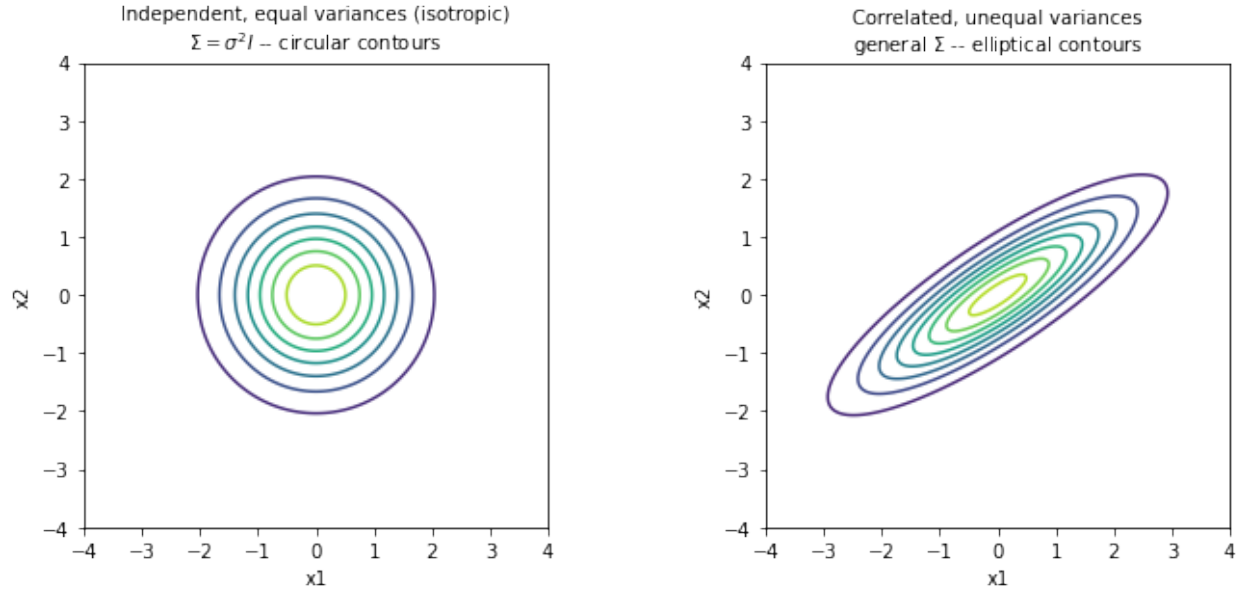
```
# independent isotropic Gaussian (circular contours) vs. a general
# (correlated / non-isotropic) Gaussian (elliptical contours) -- L7 teaser, resolved
from scipy.stats import multivariate_normal

x = np.linspace(-4, 4, 200)
y = np.linspace(-4, 4, 200)
X, Y = np.meshgrid(x, y)
pos = np.dstack((X, Y))

mu = [0, 0]
Sigma_isotropic = [[1.0, 0.0], [0.0, 1.0]]      # sigma^2 I
Sigma_general    = [[2.0, 1.2], [1.2, 1.0]]      # correlated, non-isotropic

fig, axes = plt.subplots(1, 2, figsize=(10, 4.5))
for ax, Sigma, title in zip(
    axes, [Sigma_isotropic, Sigma_general],
    ["Independent, equal variances (isotropic)\n" + r"$\Sigma=\sigma^2 I$ -- circular contours",
     "Correlated, unequal variances\n" + r"general $\Sigma$ -- elliptical contours"]):
    rv = multivariate_normal(mu, Sigma)
    ax.contour(X, Y, rv.pdf(pos), levels=8, cmap="viridis")
    ax.set_aspect("equal")
    ax.set_title(title, fontsize=10)
    ax.set_xlabel("x1"); ax.set_ylabel("x2")

plt.tight_layout()
plt.show()
```



2 Bayes' Rule

2.1 Inference

In inference, we have some unknown quantity Θ (a parameter, a hypothesis, a signal – anything unobserved) and a **prior** distribution for it that reflects our belief before seeing any data. We then observe data X , whose distribution depends on Θ . Bayes' rule produces the **posterior** distribution of Θ given the data we actually observed.

$$\underbrace{f_{\Theta}(\theta)}_{\text{prior}} \implies \text{observe } X = x \implies \underbrace{f_{\Theta|X}(\theta | x)}_{\text{posterior}}$$

Θ and X can each be discrete or continuous, giving four cases. Today we'll work through the continuous-continuous case here, then the two mixed cases in the next two sections.

2.1.1 Example: Lightbulb w/ Exponential Lifetime (estimating mean parameter)

An engineer doesn't know a bulb's failure rate λ exactly, but believes it's somewhere between 0.5 and 2.0 (per year), with no value more likely than another *a priori*:

$$\Lambda \sim \text{Uniform}(0.5, 2.0).$$

She observes one bulb's lifetime $X = x$, where $X | \Lambda = \lambda \sim \text{Exponential}(\lambda)$.

i Bayes' rule, continuous-continuous case

$$f_{\Lambda|X}(\lambda | x) = \frac{f_{X|\Lambda}(x | \lambda) f_{\Lambda}(\lambda)}{f_X(x)}, \quad f_X(x) = \int f_{X|\Lambda}(x | \lambda) f_{\Lambda}(\lambda) d\lambda.$$

```

# lightbulb: continuous prior, continuous observation
a, b = 0.5, 2.0 # prior support

def prior(lam):
    return np.where((lam >= a) & (lam <= b), 1.0/(b-a), 0.0)

def likelihood(x, lam):
    return lam * np.exp(-lam * x)

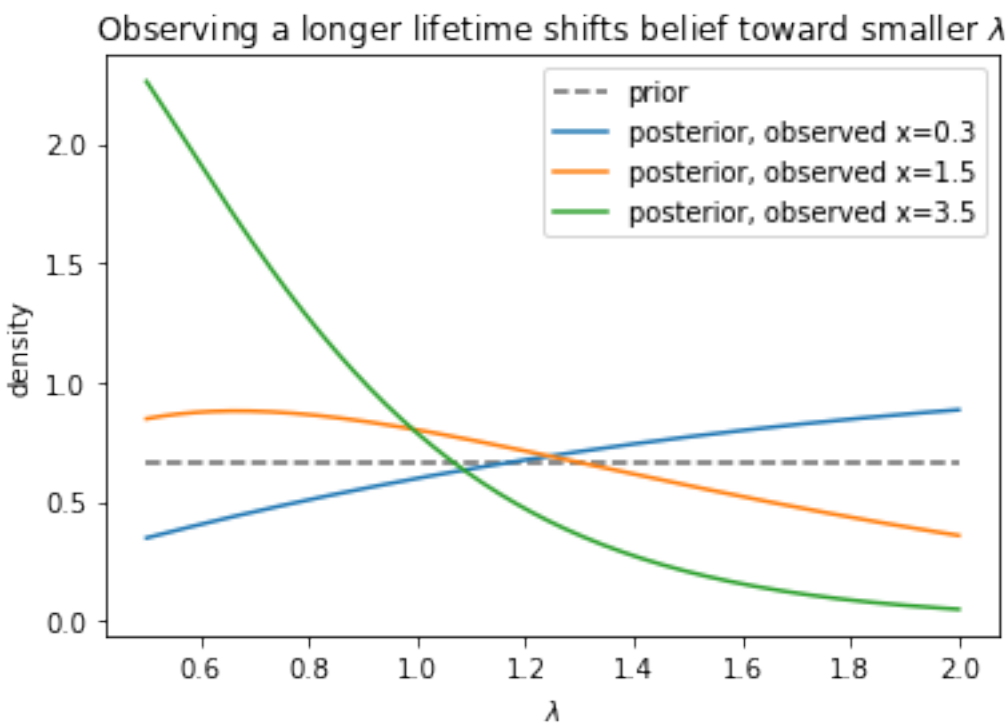
def evidence(x):
    return integrate.quad(lambda lam: likelihood(x, lam) * prior(lam), a, b)[0]

def posterior(lam, x):
    return likelihood(x, lam) * prior(lam) / evidence(x)

lams = np.linspace(a, b, 300)

fig, ax = plt.subplots()
ax.plot(lams, prior(lams), color="0.4", ls="--", label="prior")
for x_obs, color in zip([0.3, 1.5, 3.5], ["C0", "C1", "C2"]):
    post = [posterior(l, x_obs) for l in lams]
    ax.plot(lams, post, color=color, label=f"posterior, observed x={x_obs}")
ax.set_xlabel(r"$\lambda$")
ax.set_ylabel("density")
ax.legend()
ax.set_title("Observing a longer lifetime shifts belief toward smaller " + r"$\lambda$")
plt.show()

```



2.2 Inference about Discrete RV

Now suppose Θ is **discrete** but the data X we observe is **continuous**.

i Bayes' rule

Start from the definition of a conditional PDF given an event, applied to $A = \{\Theta = \theta\}$, together with the (RV form) law of total probability for f_X :

$$P(\Theta = \theta | X = x) = \frac{f_{X|\Theta}(x | \theta) P(\Theta = \theta)}{f_X(x)}, \quad f_X(x) = \sum_{\theta} f_{X|\Theta}(x | \theta) P(\Theta = \theta).$$

The denominator is computed by the Law of Total Probability from earlier in the lecture, applied with the partition $\{\Theta = \theta\}$ over all possible values of θ .

2.2.1 Example: Signal Detection

Recall our previous signal-detection setup: a binary signal $S \in \{-1, +1\}$ is sent with equal prior probability, through a channel with additive noise $N \sim N(0, \sigma^2)$ independent of S , so we observe $Y = S + N$. We previously derived the *error probability* of the rule “decide +1 if $Y > 0$ ”; today we derive that rule itself, as a consequence of Bayes' rule.

By the formula above (with $\Theta = S$, discrete; $X = Y$, continuous):

$$P(S = +1 | Y = y) = \frac{\phi\left(\frac{y-1}{\sigma}\right) \cdot \frac{1}{2}}{\phi\left(\frac{y-1}{\sigma}\right) \frac{1}{2} + \phi\left(\frac{y+1}{\sigma}\right) \frac{1}{2}} = \frac{1}{1 + e^{-2y/\sigma^2}},$$

where ϕ is the standard normal density.

💡 Recovering the decision rule

$P(S = +1 | Y = y) > \frac{1}{2}$ exactly when $y > 0$. The rule that we evaluated failure probability for previously is the rule that picks the **more probable** signal given the observation – this is called the *maximum a posteriori (MAP)* rule.

2.3 Inference Based on Discrete Observations

Suppose now Θ may be **continuous** while the data N we observe is **discrete**.

i Bayes' rule, continuous Θ / discrete N

$$f_{\Theta|N}(\theta | n) = \frac{P(N = n | \Theta = \theta) f_{\Theta}(\theta)}{P(N = n)}, \quad P(N = n) = \int P(N = n | \Theta = \theta) f_{\Theta}(\theta) d\theta.$$

2.3.1 Example: Estimating a Coin's Bias

A coin's probability of heads Θ is unknown; take a totally uninformative prior $\Theta \sim \text{Uniform}(0, 1)$ — every possible bias is equally plausible before you flip. Flip it n times and observe $K =$ number of heads, where $K | \Theta = \theta \sim \text{Binomial}(n, \theta)$.

Plugging the Binomial likelihood and the (constant) uniform prior density $f_{\Theta}(\theta) = 1$ into the continuous- Θ /discrete-observation form of Bayes' rule from the previous slide:

$$f_{\Theta|K}(\theta | k) = \frac{\binom{n}{k} \theta^k (1-\theta)^{n-k} \cdot 1}{P(K=k)} \propto \theta^k (1-\theta)^{n-k}, \quad 0 \leq \theta \leq 1.$$



The “ $\propto$ ” symbol means “proportional to” and drops any θ -free factor (ignoring the normalization factors since they don't depend on θ).

2.3.2 The Posterior Is a Named Distribution

Normalizing means dividing by $\int_0^1 \theta^k (1-\theta)^{n-k} d\theta$, which has the closed form $\frac{k!(n-k)!}{(n+1)!}$ (a Beta-function value at integer arguments). So

$$f_{\Theta|K}(\theta | k) = \frac{\theta^k (1-\theta)^{n-k}}{k!(n-k)!/(n+1)!}, \quad 0 \leq \theta \leq 1,$$

which is exactly the density of a $\text{Beta}(k+1, n-k+1)$ random variable.

i Two different point estimates from the same posterior

The **mode** of $\text{Beta}(k+1, n-k+1)$ — the value of θ that maximizes the posterior density, i.e. the *MAP estimate* — is

$$\hat{\theta}_{\text{MAP}} = \frac{k}{n},$$

exactly the naive sample proportion. The **mean** of the same distribution — the Bayes estimator under squared-error loss — is

$$\hat{\theta}_{\text{mean}} = \frac{k+1}{n+2},$$

the classical **Laplace's rule of succession** (add one hypothetical success and one hypothetical failure, then average).

```
# coin-bias inference: posterior concentrates as n grows (same observed proportion)
theta_true = 0.65
theta_grid = np.linspace(0, 1, 400)

def unnorm_posterior(theta, n, k):
    return theta**k * (1-theta)**(n-k)

fig, ax = plt.subplots()
for n in [5, 20, 100, 500]:
    k = int(round(theta_true * n)) # keep the observed proportion k/n ~ theta_true
    up = unnorm_posterior(theta_grid, n, k)
    Z = integrate.quad(lambda t: unnorm_posterior(t, n, k), 0, 1)[0]
    ax.plot(theta_grid, up/Z, label=f"n={n}, k={k}")
```

```

ax.axvline(theta_true, color="k", lw=1, ls="--", label=r"true  $\theta$ ")
ax.set_xlabel(r" $\theta$ "); ax.set_ylabel("posterior density")
ax.legend()
ax.set_title("More flips (same observed fraction) -> sharper posterior")
plt.show()

```

